Reducing Negative Transfer in Universal Anomaly Detection via Reliability-Aware Meta-Routing

(working skeleton — empirics pending)

Abstract

No single anomaly detector dominates across domains: a method that is strong on tabular data is often weak on time series, vision, or out-of-distribution detection, and naively transferring or fusing detectors causes *negative transfer*. We ask whether a single, domain-agnostic *meta-router* can reduce that negative transfer using only validation-derived reliability signals. The router consumes per-detector anomaly scores, confidence and calibration features, detector disagreement, modality availability, and dataset-level meta-features, and either selects a detector, chooses a fusion rule, or abstains. We do *not* claim to be best on every dataset—ADBench shows no method can be. We test a sharper, falsifiable claim: that reliability-aware meta-routing improves **average rank**, raises **win-rate** against strong baselines, and lowers **worst-case regret** and **negative-transfer rate** across heterogeneous anomaly benchmarks (tabular, time series, vision/OOD, industrial, cyber/fraud), with no domain-specific test tuning and bootstrap CIs above zero. [Empirical results pending; this skeleton fixes the claim, the protocol, and the falsification criteria *before* any benchmark is scored.]

1 Claim and Falsification (pre-registered)

Primary claim. A reliability-aware meta-router R achieves, across a suite of K benchmark datasets spanning ≥ 4 domains, strictly better *average rank* than (i) the best fixed single detector family and (ii) a strong meta-baseline (MetaOD-style selection), with a paired bootstrap 95% CI on the rank improvement that excludes zero.

Secondary endpoints. (a) win-rate > 0.5 vs each baseline family; (b) reduced worst-case (max over datasets) regret to the per-dataset oracle; (c) reduced negative-transfer rate (fraction of datasets where the routed choice underperforms a robust default such as ECOD); (d) an ablation showing the reliability features are necessary (router with reliability features beats router without).

Falsification. If R does not beat a robust fixed default (ECOD / Isolation-Forest) on average rank with CI>0, the universal claim FAILS and is reported as a negative result. No post-hoc dataset selection is permitted.

2 Gate U: Universal Cross-Domain Reliability (the pass contract)

Replaces the old (clean-transfer) Gate E. Pass requires ALL of:

1. average rank better than the best single baseline family AND a meta-baseline;

2. win-rate above strong baselines across domains;
3. worst-case regret reduced vs every baseline;
4. negative-transfer rate reduced vs a robust default;
5. no domain-specific test tuning (selection on validation/meta-features only);
6. paired bootstrap CI on the rank gain > 0, pooled across dataset families;
7. ablation: reliability router strictly beats the no-reliability router.

3 Method: ELARA as a Meta-Router

Inputs (per dataset, per sample where applicable): anomaly scores from a detector zoo; per-detector confidence; validation calibration error (ECE); score entropy/sharpness; pairwise detector disagreement; modality availability mask; dataset meta-features (size, dimensionality, contamination estimate, feature-type mix). **Outputs:** (a) select a detector, (b) select a fusion rule, (c) abstain / fail-closed, (d) a calibrated anomaly score. All routing parameters are fit on validation/meta-features only; test labels are never used for selection. The degenerate-channel guard (this project) is folded in as the floor so no inverted/saturated detector can drive the routed score.

4 Benchmark Suite

Domain	Source
Tabular	ADBench (57 datasets)
Time series	TimeEval / TSB-AD
Vision / OOD	OpenOOD, OODDB
Industrial vision	MVTec AD 2, VisA, Real-IAD, Real-IAD D3
Cyber / fraud	UNSW-NB15, CIC-IDS, credit-card / IEEE-CIS fraud

5 Detector Zoo (base experts)

Isolation Forest, LOF, ECOD, COPOD, OCSVM, DeepSVDD (tabular/general); PatchCore, EfficientAD, DINO memory-bank (industrial vision); standard time-series and OOD baselines. All run once per dataset; scores archived in a common format.

6 Statistics

Average rank (Friedman/critical-difference), win-rate, regret to per-dataset oracle, negative-transfer rate, paired bootstrap CIs pooled across families, Holm-corrected across the family of endpoints.

7 Status

This is a pre-registered skeleton. The empirical claim is unverified until the suite is scored. The disciplined build order is: (1) tabular pilot (ADBench subset) to get a first honest average-rank signal vs ECOD/IForest/MetaOD; only if the router wins there do we (2) add the remaining domains and (3) write the full paper. If the pilot fails, this is reported as a negative result, not buried.